

Formula Sheet- Field Units for Well Control Drilling Operations



(361) 575-4882

Measurement	Units	Decimals	DRILLING IADC Rounding Rules Kill mud round <u>up</u> 10.11 = 10.2 & 11.03 = 11.1 Leak off mud round <u>down</u> 10.16 = 10.1 x = Whole numbers x . x = Number with 1 decimal x . x x x x = Number with 4 decimals
Depth	ft	x	
Pressure	psi	x	
Pressure Gradient	psi/ft	x . x x x x	
Mud Weight	ppg	x . x	
Volume	bbls	x . x	
Capacity & Displacement	bbls/ft	x . x x x x	
Pump speed in strokes per minute	SPM	x	
Strokes	stk or stks	x	
Speed in feet per hour	ft/hr	x	
Area	in ²	x . x x x x	
Force	lbs	x	
ICP to FCP = DP pressure reduction surf to bit	psi/10 steps or psi/100 stks	x	

PEMDAS order of mathematical operations = Parenthesis then Exponents then Multiplication/Division then Addition/Subtraction

Pressures - PSI

Pressure psi =	Force lbs ÷ Diameter ² ÷ 0.7854	(Diameter in Inches)
Hydrostatic Pressure Gradient psi/ft =	Mud Weight ppg × 0.052	
Hydrostatic Pressure psi =	Mud Weight ppg × 0.052 × TVD ft	
Formation Pressure psi or Bottom Hole Pressure at Shut in =	Hydrostatic pressure in drill string psi + SIDPP psi	
Initial Circulating Pressure psi =	Slow Circulating Rate Pressure psi + SIDPP psi	
Final Circulating Pressure psi =	Slow Circulating Rate Pressure psi × (Kill Mud Weight ppg ÷ Original Mud Weight ppg)	
MAASP or MACP psi =	(Maximum Allowable Mud Weight ppg - Current Mud Weight ppg) × 0.052 × Shoe TVD ft	
New MAASP after Kill psi =	(Maximum Allowable Mud Weight ppg - Kill Mud Weight ppg) × 0.052 × Shoe TVD ft	
New Pump Pressure with New SPM psi =	Current Pressure psi × (New SPM ÷ Old SPM) ²	(only approximate)
New Pump Pressure with New Mud Weight psi =	Current Pressure psi × (New Mud Weight ppg ÷ Old Mud Weight ppg)	(only approximate)
Pressure per Barrel psi/bbl =	Fluid Gradient psi/ft ÷ Annular Cap bbls/ft at Top of Well	
or	(Fluid Gradient psi/ft - Gas Gradient (0.104) psi/ft) ÷ Annular Cap bbls/ft at Top of Well	
Subsea Casing or Choke Pressure after pump Start Up psi =	SICP psi - Choke Line Friction Loss/pressure psi	

Pounds per Gallon - PPG

Mud Weight ppg =	Pressure psi ÷ TVD ft ÷ 0.052 or Pressure Gradient psi/ft ÷ 0.052	
Equivalent Mud Weight ppg =	Pressure psi ÷ 0.052 ÷ TVD ft or (Surface Pressure psi ÷ TVD ft ÷ 0.052) + Mud Weight ppg	
Equivalent Circulating Density ppg =	(Annular Pressure Loss psi ÷ 0.052 ÷ TVD ft) + Original Mud Weight ppg	
Kill Mud Weight ppg =	(SIDPP psi ÷ 0.052 ÷ TVD ft) + Original Mud Weight ppg	
Maximum Allowable Mud Weight or Fracture Mud ppg =	(Surface Leak Off psi ÷ 0.052 ÷ Shoe TVD ft) + Test Mud Weight ppg	
Subsea Riser Margin ppg =	[(Riser Mud HP psi - Seawater HP psi) ÷ 0.052] ÷ (Well TVD ft - Water Depth ft - Air Gap ft)	

Barrels per foot - bbls/ft or Barrels - bbls

Tubular Capacity bbls/ft =	ID ² ÷ 1029.4	(ID = Inside Diameter of Tubular)
Annular Capacity bbls/ft =	(ID of Casing ² or Hole Size ² - OD of Tubing ²) ÷ 1029.4	(OD = outside diameter of Tubular)
Volume to Bleed due to Gas Migration bbls =	(Working Pressure to Bleed psi ÷ Mud Gradient psi/ft) × Annular Cap bbls/ft at Top of Well	
Additional Mud Return by Slug bbls =	[(Slug Weight ppg ÷ Mud Weight ppg) - 1] × Slug Volume bbls	
Total Mud Returned by Slug bbls =	(Slug Weight ppg ÷ Mud Weight ppg) × Slug Volume bbls	
Rectangular Tank bbls/ft =	(Length ft × Width ft) ÷ 5.61	(To convert to Inches) ÷ by 12 = bbls/inch

Feet - FT

Level Drop after Pumping a Slug ft =	[(Slug Weight ppg ÷ Mud Weight ppg) - 1] × Slug Vol bbls ÷ Drill Pipe Cap bbls/ft	
Gas Migration Rate ft/hr =	Shut in Pressure Increase psi/hr ÷ Mud Gradient psi/ft	(can use SIDPP or SICP)
Height of Fluid in a Pipe or Annulus ft =	Kick Volume bbls ÷ Annular Capacity bbls/ft or Pipe Capacity bbls/ft	

Miscellaneous

Boyle's Law Formulae =	P1 × V1 = P2 × V2 or P2 = (P1 × V1) ÷ V2 or V2 = (P1 × V1) ÷ P2	
	P = pressure V = volume Atmospheric Pressure = 14.7 psi	
Force lbs =	Pressure psi × Diameter ² × 0.7854	(Diameter in Inches)

Formula Sheet- Field Units for Well Control Workover Operations



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Pressures - PSI

WORKOVER
COMPLETION
WELL SERVICING

$$\text{Pressure psi} = \text{Force lbs} \div \text{Diameter}^2 \div 0.7854$$

$$\text{Hydrostatic Pressure Gradient psi/ft} = \text{Mud Weight ppg} \times 0.052$$

$$\text{Hydrostatic Pressure psi} = \text{Mud Weight ppg} \times 0.052 \times \text{TVD ft}$$

$$\text{Formation or Bottom Hole Pressure at Shut in psi} = \text{Hydrostatic Pressure in Production String psi} + \text{Shut in Surface Pressure psi}$$

$$\text{Formation Fracture Pressure psi} = \text{Formation Fracture Gradient psi/ft} \times \text{Top Perforations TVD ft}$$

$$\text{Initial Hydrostatic Pressure psi} = \text{Formation Pressure psi} - \text{Shut in Tubing Pressure psi}$$

$$\text{Initial Average Fluid Density ppg} = \text{Initial Hydrostatic Pressure psi} \div \text{Top Perforations TVD ft} \div 0.052$$

$$\text{Maximum Initial Surface Pressure psi} = \text{Formation Fracture Pressure psi} - \text{Initial Hydrostatic Pressure psi}$$

$$\text{Maximum Final Surface Pressure psi} = \text{Formation Fracture Pressure psi} - (\text{Kill Fluid Weight ppg} \times 0.052 \times \text{Top Perforations TVD ft})$$

$$\text{Pressure per Barrel psi/bbl} = \text{Fluid Gradient psi/ft} \div \text{Annular Capacity bbls/ft at Top of Well}$$

$$\text{or} \quad (\text{Fluid Gradient psi/ft} - \text{Gas Gradient (0.104) psi/ft}) \div \text{Annular Capacity bbls/ft at Top of Well}$$

Pounds per Gallon - PPG

$$\text{Fluid Weight ppg} = \text{Pressure psi} \div \text{TVD ft} \div 0.052 \quad \text{or} \quad \text{Pressure Gradient psi/ft} \div 0.052$$

$$\text{Equivalent Fluid Weight ppg} = \text{Pressure psi} \div 0.052 \div \text{TVD ft} \quad \text{or} \quad (\text{Surface Pressure psi} \div \text{TVD ft} \div 0.052) + \text{Fluid Weight ppg}$$

$$\text{Kill Fluid Weight ppg} = (\text{Shut in Tubing Pressure psi} \div 0.052 \div \text{Top Perforations TVD ft}) + \text{Original Fluid Weight ppg}$$

$$\text{or Kill Fluid Weight ppg} = (\text{Bottom Hole Pressure psi} \div 0.052 \div \text{TVD ft})$$

$$\text{Fluid Density to Mix ppg} = \text{Fluid Density at Avg Temp ppg} + [(\text{Avg Temp } F - \text{Surface Temp } F) \times \text{Weight Loss ppg/degree } F]$$

Avg = Average F = Degrees Fahrenheit

Example Weight Loss Chart: (Values will vary based on type of fluid & other factors)

Example: Brine Weight Weight Loss (ppg / °F)

8.4 - 9.0 0.0017

9.1 - 11.0 0.0025

Barrels per foot - bbls/ft or Barrels bbls:

$$\text{Tubular Capacity bbls/ft} = \text{ID}^2 \div 1029.4 \quad (\text{ID} = \text{Inside Diameter of Tubular})$$

$$\text{Annular Capacity bbls/ft} = (\text{ID of Casing}^2 \quad \text{or} \quad \text{Hole Size}^2 - \text{OD of Tubing}^2) \div 1029.4 \quad (\text{OD} = \text{Outside Diameter of Tubular})$$

$$\text{Volume to Bullhead bbls} = \text{Surf lines bbls} + \text{Surf to EOT bbls} + \text{EOT to Top Perfs bbls} + \text{Top Perfs to Bottom Perfs bbls} + (\text{Over Disp. bbls})$$

$$\text{Volume Bleed due to Gas Migration bbls} = (\text{Working Pressure to Bleed psi} \div \text{Mud Gradient psi/ft}) \times \text{Annular Cap bbls/ft at Top of Well}$$

$$\text{Additional Fluid Return by Slug bbls} = [(\text{Slug Weight ppg} \div \text{Fluid Weight ppg}) - 1] \times \text{Slug Volume bbls}$$

$$\text{Total Fluid Returned by Slug bbls} = (\text{Slug Weight ppg} \div \text{Fluid Weight ppg}) \times \text{Slug Volume bbls}$$

$$\text{Rectangular Tank bbls/ft} = (\text{Length ft} \times \text{Width ft}) \div 5.61 \quad (\text{To convert to inches}) \div \text{by 12} = \text{bbls/inch}$$

Feet - FT

$$\text{Level Drop after Pumping a Slug ft} = [(\text{Slug Weight ppg} \div \text{Fluid Weight ppg}) - 1] \times \text{Slug Volume bbls} \div \text{Tubular Cap bbls/ft}$$

$$\text{Gas Migration Rate ft/hr} = \text{Shut in Pressure Increase psi/hr} \div \text{Mud Gradient psi/ft} \quad (\text{use SIDPP or SICP})$$

Miscellaneous

$$\text{Bullhead SPM to Exceed Gas Migration} = (\text{Gas Migration rate ft/hr} \div 60) \times \text{Tubing Cap bbls/ft} \div \text{Pump output bbls/stk}$$

$$\text{Boyle's Law Formulae} = P_1 \times V_1 = P_2 \times V_2 \quad \text{or} \quad P_2 = (P_1 \times V_1) \div V_2 \quad \text{or} \quad V_2 = (P_1 \times V_1) \div P_2$$

P = pressure V = Volume Atmospheric Pressure = 14.7 psi

$$\text{Force lbs} = \text{Pressure psi} \times \text{Diameter}^2 \times 0.7854 \quad (\text{Diameter in Inches})$$

Well Control Formula Constants & Abbreviations

$$0.052 = \text{Factor converting weight to psi/ft} \quad 1029.4 = \text{Conversion to capacity bbls/ft} \quad 5.61 \text{ cubic ft} = 1 \text{ bbl} \quad \% \text{ EFF} = \text{Percent efficiency}$$

$$8.33 = \text{Weight of fresh water} \quad \text{Stks} = \text{strokes} \quad \text{Strokes per minute} = \text{stks/min} \quad \text{Slow Circulating Rate Pressure} = \text{SCRPP}$$

$$\text{Barrels} = \text{bbls} \quad \text{Barrels per foot} = \text{bbls/ft} \quad \text{Barrels per stroke} = \text{bbls/stk} \quad \text{Barrels per minute} = \text{bbls/min}$$

$$\text{Casing} = \text{Csg} \quad \text{Drill pipe} = \text{DP} \quad \text{Drill collars} = \text{DC} \quad \text{Heavy weight drill pipe} = \text{HWDP} \quad \text{Open hole} = \text{OH} \quad \text{End of tubing} = \text{EOT}$$

$$\text{Feet} = \text{ft} \quad \text{IN} = \text{inch} \quad \text{Measured depth} = \text{MD} \quad \text{True vertical depth} = \text{TVD} \quad -1 = \text{correction constant for additional slug/volume (air in pipe)}$$

$$\text{Shut in drill pipe pressure} = \text{SIDPP} \quad \text{Shut in casing pressure} = \text{SICP} \quad \text{Shut in tubing pressure} = \text{SITP} \quad \text{Perforations} = \text{Perfs}$$

$$\text{Inside diameter} \times \text{inside diameter} = \text{ID}^2 \quad \text{Outside diameter} \times \text{outside diameter} = \text{OD}^2 \quad \text{Open hole} \times \text{open hole} = \text{OH}^2$$

$$35 = \text{Weight of 1 gallon of barite} \quad \text{Sacks} = \text{SXS} \quad \text{Sacks per barrel} = \text{sxs/bbl} \quad \text{Sacks per minute} = \text{sxs/min}$$

$$\text{Initial circulating pressure} = \text{ICP} \quad \text{Final circulating pressure} = \text{FCP} \quad \text{Subsea} = \text{Choke line friction loss/pressure} = \text{CLFP}$$